

AUDITORY GROUPING DAILY

Friday, 10 July 2026 · 412 papers screened · curated

Canon day, so the deep read is Bregman rather than a new paper — fitting, since today's automated haul was thin on chunking, sequences, or auditory-prefrontal circuits, so most of what follows came from searching outside the candidate pool. The standout is a bioRxiv preprint from Christine Constantinople's lab: an orbitofrontal-to-caudoputamen circuit that updates a rat's latent belief about a hidden reward state as an evolving population distribution, a close cousin of your GLM Module C slow-drift framing. An eLife mouse-mPFC paper is worth reading against your prelimbic recordings, since it shows much apparent mPFC encoding is a movement/arousal confound that survives even when silencing barely touches behavior. A bioRxiv preprint from Sahar Moghimi's group pushes metrical-grouping sensitivity down into the newborn period, a nice bookend to today's Bregman reading. Given how sparse the real hits were, this digest stays short rather than padding out weaker candidates.

CHUNKING & SEQUENCE PROCESSING

Effect of tempo on newborns' neural processing of auditory rhythm: Emergence of sensitivity to metrical structure

BIORxIV Edalati, M. et al. — bioRxiv — 2026-07-08

Corresponding: Sahar Moghimi — University of Picardie Jules Verne — France | [Open paper](#)

Newborns show neural sensitivity not just to a repeating auditory beat but to the slower beat-grouping periodicities (pairs or triples of beats forming a metrical hierarchy), and critically this hierarchical sensitivity only emerges at some tempi and not others, suggesting the immature auditory system has tempo-dependent limits on how much temporal structure it can extract at once. This is a direct developmental precedent for your own chord-sequence-in-stream work: your target sequence is itself a small hierarchy (a fixed 3-4 element chunk that must be detected regardless of surrounding context), and this paper's tempo-dependent breakdown of grouping sensitivity is a concrete reason to ask whether your own chord duration and inter-chord interval (150 ms/150 ms) sit inside or outside the window where a naive listener (or a mouse early in training) could extract hierarchical structure at all before learning kicks in.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Medial prefrontal cortex encodes but is not required to generate goal-directed actions under threat.

PUBMED Muhammad S Sajid et al. — eLife — 2026

Corresponding: Manuel A. Castro-Alamancos — University of Connecticut — United States | [Open paper](#)

Using fiber photometry, single-cell calcium imaging, and optogenetic inhibition in mouse medial prefrontal cortex during a learned avoidance task, the authors show that a large share of apparent task-related activity in this region is actually shared with a control, non-task cortical area once movement and arousal are properly regressed out, and that silencing the mPFC GABAergic neurons with the cleanest encoding of demanding avoidance contingencies had almost no effect on learning or performance. This is a direct methodological caution for your prelimbic cortex recordings during the mouse sequence-discrimination task: it argues for validating any decision-window encoding you find in prelimbic cortex against a control region and, where possible, an optogenetic perturbation, before treating anticipatory discrimination signals as causally load-bearing rather than a correlate of behavioral state.

BEHAVIORAL & COMPUTATIONAL MODELING

A corticostriatal circuit updates subjective beliefs about latent task states

CROSSREF (MODEL-DISCOVERED) Margaret L. DeMaegd et al. — 2026-03-14

Corresponding: Christine M. Constantinople — New York University — United States | [Open paper](#)

Projection-specific recordings and optogenetic perturbations from orbitofrontal neurons projecting to the caudoputamen in rats performing a hidden-reward-state task show that this corticostriatal pathway encodes categorical evidence for a latent state and that downstream neurons could in principle decode the animal's full belief distribution over states as it deliberates, giving a circuit-level implementation of belief updating about an abstract, non-observable variable. This is the closest circuit-level precedent I've seen for what your own GLM Module C (slow drifts in strategy from satiety or time-on-task) is trying to capture behaviorally: it suggests a concrete corticostriatal locus you could look to, or at least borrow the belief-decoding logic from, if you ever want to move Module C from a behavioral drift term into a claim about what a specific circuit is representing.

CIRCUITS, METHODS & POPULATION ANALYSIS

Visual working memory guides attention rhythmically in humans.

PUBMED Jiachen Lu et al. — eLife — 2026

Corresponding: Xilin Zhang — South China Normal University — China | [Open paper](#)

Rather than one or many working-memory items being simultaneously available as an attentional template, this paper finds that two remembered items alternately dominate behavioral guidance at a theta rhythm (4-8 Hz), with frontal theta phase leading the occipital alpha oscillation that tracks which item is currently prioritized. If template-guided attention is intrinsically rhythmic rather than continuously available, it's worth asking whether hit/miss timing in your human chunk-in-stream detection task shows periodicity locked to either the stimulus tempo or an endogenous rhythm, which would suggest the memorized target template is only intermittently 'online' for comparison against the incoming stream rather than being a static comparison signal.

OTHER NOTABLE

C1q and immunoglobulins mediate activity-dependent synapse loss in the adult brain

RSS:SCIENCE Science — 2026-07-09

Corresponding: Beth Stevens — Harvard Medical School — United States | [Open paper](#)

This Science paper (with an accompanying perspective piece, 'Antibodies sculpt adult brain circuits') shows that neuronal hyperactivity triggers region-specific deposition of complement protein C1q, and that B-lymphocyte-lineage cells and immunoglobulins are required for this activity-dependent synapse elimination in the adult hippocampus, extending complement-mediated pruning (previously known mainly from development and disease) into ongoing, activity-driven circuit refinement in the healthy adult brain. It sits well outside your auditory/sequence paradigm, but it is a field-defining mechanistic result about how immune signaling gates activity-dependent circuit remodeling, which is the same general class of question (what determines which synapses survive experience-driven plasticity) that sits underneath the auditory-cortex critical-period literature already in your canon list.

Bregman (1990), **Auditory Scene Analysis** — foundational framework (cite a review chapter if the book isn't fetchable).

Foundations pick — abstract only

- Using fMRI and computational modeling, this group formally dissociates two distinct triggers for perceived event boundaries in continuous experience: high prediction error (mismatch between expected and observed information) and high prediction uncertainty (reduced confidence in the prediction itself) — accounts that prior work had largely conflated into one 'boundary' signal.
- The two boundary types produce distinguishable spatiotemporal signatures: error-driven boundaries evoke early pattern shifts in ventrolateral prefrontal areas followed by pattern stabilization in prefrontal and temporal areas, while uncertainty-driven boundaries evoke shifts confined to parietal regions of the dorsal attention network with minimal subsequent stabilization.
- Across both boundary types there is a conserved temporal cascade around human-identified boundaries — early pattern shifts in anterior temporal regions (about 11.9 s before the boundary), followed by shifts in parietal areas (about 4.5 s before), then whole-brain pattern stabilization (about 11.8 s after) — suggesting event segmentation is a distributed, sequenced process rather than a single mechanism firing at one moment.
- This gives chunking a genuine operational, two-factor definition (error versus uncertainty) with dissociable neural signatures and timing, rather than treating 'a boundary was detected' as a single undifferentiated event — directly useful for thinking about what a chunk boundary in an auditory stream actually is, computationally, not just perceptually.

HOW TO READ IT (≈20 min)

About 20 minutes, working from the abstract and title alone (the automated full-text fetch for this paper failed and returned unrelated content, so treat anything beyond the abstract as unverified until you pull the actual eLife PDF yourself — doi.org/10.7554/eLife.107955): start by re-reading the abstract slowly twice, since nearly every clause carries a distinct claim (two boundary-generating models, three region-by-timing dissociations, one shared cascade). When you get the real paper, prioritize the Results section describing the error-driven versus uncertainty-driven pattern-shift regions and their timing, and the Methods description of how the two computational models actually generate a 'boundary' from prediction error versus prediction uncertainty — that computational operationalization is the part most transferable to your own modeling toolchain. Skim the Introduction for how this reconciles with older event-segmentation-theory accounts, and skip any methods details specific to fMRI preprocessing/denoising unless you're troubleshooting your own imaging pipeline.

YOUR QUESTION TO ANSWER

In the human chunk-in-stream task, a Start-Target or End-Target trial gives you a natural candidate boundary — the transition into or out of the memorized template. If you tagged each such transition as more error-like (a template match violated by an unexpected element) or more uncertainty-like (a genuinely ambiguous, near-miss distractor) using this paper's operational definitions, would you predict your high-gamma responses at those transitions to show the same anterior-temporal-then-parietal cascade — and which of your five staged difficulty levels would give you the cleanest contrast to test whether that cascade's timing changes as the task gets harder?